INCH POUND
MIL-PRF-1/29D
1 July 1997
SUPERSEDING
MIL-E-1/29C
19 February 1970

PERFORMANCE SPECIFICATION SHEET
ELECTRON TUBE, RECEIVING
TYPE 1Z2

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-PRF-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Rectifier, half wave, high vacuum

Outline Base Envelope	See figure 1

Cathode - - - Thoriated tungsten

Base connections:

Pin No.	1	2	3	4	5	6	7	Cap
Element	-f	+f	-f	-f	+f	-f	+f	a
	int sd		int sd	int sd		int sd		

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	epx	ib	Io	CL	Rl	F1
Unit:	Vac	kv	ma	mAdc	µF	Meg	kHz
Maximum	1.25 ± 5%	15	8.5	1.5	- - -	- - -	200
Minimum:	- - -	- - -	- - -	- - -	- - -	(see note 1)	- - -
Test Conditions:	1.25	15	- - -	1.5	0.01	4.2	- - -

GENERAL:

Qualification - Not required

DISTRIBUTION STATEMENT A.Approved for public release; distribution is unlimited.

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Table 1. Testing and Inspection.

Method	Requirement or test	Conditions	Acceptance Level (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1							
1353	Operation of rectifiers	Ef =1.2 V; Zp/ib = 8.5 ma (min) (see notes 2 and 3)	0.65	II	- - -	- - -	- - -	- - -
1231	Emission current	Ef = 1.35V: Eb = 100 V dc	0.65	II	Is	9.5	- - -	mA dc
	Quality conformance inspection, part 2							
1301	Filament current		- - -	- - -	If	245	285	mA
1031	Low-frequency vibration	No voltages	- - -	- - -	- - -	- - -	- - -	- - -
1101	Secureness of base, cap, or insert		- - -	- - -	- - -	- - -	- - -	- - -
1105	Permanence of marking		- - -	- - -	- - -	- - -	- - -	- - -
	Quality conformance inspection, part 3							
1501	Intermittent life	Group A; 250 hours with Ef = 1.25 V, or 500 hours with Ef=1.2V; Zp/ib = 8.5 ma	- - -	- - -	- - -	- - -	- - -	- - -
- - -	Intermittent life-test end point	Failure to rectify and emission	- - -	- - -	Is	8.5	- - -	mA dc

NOTES:

1/ The supply source impedance shall be at least 300,000 ohms.

2/ The tube shall be operated in a half-wave rectifier circuit at 50 to 60 Hz. Apply heater and anode voltage simultaneously to cold tube. After cathode has reached operating temperature, key anode current three times, no arcing shall occur.

3/ This test to be performed at the conclusion of the holding period.

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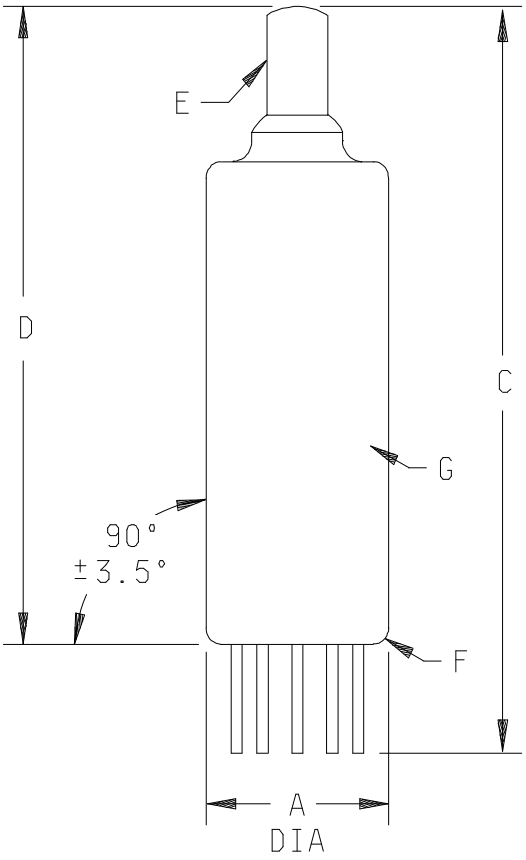

Dimensions in inches with metric equivalents (mm) in parentheses		
Ltr	Minimum	Maximum
Quality conformance inspection, part 1		
A		.75 (19.05)
C		2.70 (68.58)
D	2.34 (59.44)	2.47 (62.74)
Quality conformance inspection, part 3 (periodic check)		
E	Cap: C1-3	
F	Base: E7-1	
G	Envelope: T5½	

FIGURE 1. Outline drawing of electron tube type 1Z2.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - ME
Navy - AS, CG, OS
Air Force - 11, 19, 99

Preparing activity:
DLA - CC

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